

Pavan Kalyan Hanumanthu

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Professional Summary:

Full Stack Engineer with over 5+ years of experience architecting and implementing scalable, secure, and reliable backend solutions for diagnostic and analytics platforms in high-impact Agile environments. Adept at building cloud-native microservices and distributed systems, with advanced expertise in Java, Python, Spring Boot, AWS, Docker, and Kubernetes.

Specialized in diagnostic tooling for service and manufacturing, improving simplicity and reducing manual intervention through proactive automation and system observability. Strong proponent of CI/CD, DevOps best practices and automated release processes.

Demonstrates leadership through mentoring, technical documentation, and cross-team collaboration to deliver robust software with high quality and compliance standards. Experienced in integrating generative AI solutions and enhancing backend platforms to support continuous improvement and expansion of diagnostic capabilities.

Excellent communicator who bridges the gap between engineering, product management and quality teams for holistic solutions.

Technical Skills:

- **Languages:** Java, Python, JavaScript, SQL, C
- **Frontend:** Angular, React, HTML5, CSS3, Bootstrap
- **Frameworks & Libraries:** Spring Boot, Spring MVC, REST APIs, JPA, Hibernate, NodeJS
- **Cloud & DevOps:** AWS, GCP, Azure, Docker, Kubernetes, Jenkins, GitHub Actions, Maven, Gradle, Terraform
- **Messaging & Database:** Kafka, RabbitMQ, MongoDB, Oracle DB, PostgreSQL, MySQL
- **Monitoring & Testing:** JUnit, Mockito, Cypress, Postman, CloudWatch, ELK Stack, Prometheus
- **API Integration:** OAuth 2.0, JWT, SAML, SSO, API Gateway
- **SDLC:** Agile/Scrum, TDD, DevSecOps, Code Reviews, CI/CD
- **Tools:** Swagger, OpenAPI, SVN, SourceTree
- **Observability:** Automated diagnostics, distributed tracing, log aggregation

Education:

Master’s in Computer Science (GPA: 3.7)
Wright State University, Dayton, OH

Aug 2023 – May 2025

Certificates:

Computer Programming (Vector India Pvt Ltd) - Mar 2020
Java Development (Udemy)

Professional Experience:

Wright State University – Dayton, OH

Software Developer

Apr 2024 – May 2025

- Upgraded Angular from v14 to v18, improving modularity and decreasing page load time by 40% for research and analytics web applications.
- Led the development of more than 15 microservices in Java, Spring Boot, and Hibernate, boosting fault tolerance and system observability for machine learning diagnostic pipelines.
- Designed and maintained REST APIs using Swagger/OpenAPI for seamless integration between frontend dashboards and backend data analytics tools.
- Automated infrastructure and release management with Jenkins, Docker, Kubernetes, and multi-cloud deployments, ensuring high reliability and continuous integration.
- Implemented RBAC and OAuth2 security, maintaining compliance and eliminating unauthorized access incidents.
- Created a reusable React and TypeScript UI component library, accelerating feature rollout across modules and standardizing application design.
- Facilitated cross-functional collaboration and mentored junior developers through regular peer code reviews and technical sessions.
- Enhanced diagnostic features by integrating error-logging, device compatibility analytics, and real-time monitoring for research platforms.

HTC Global Services – India

Full Stack Developer

Mar 2021 – Jul 2023

- Engineered over 25 Angular UI components, leveraging lazy loading, RxJS observables, and modular architecture for faster client-side rendering and improved dynamic interactions.
- Built backend microservices with Spring Boot, Maven, and RESTful APIs serving highly available and secure transactional platforms.
- Automated release cycles and infrastructure deployment through Jenkins, Docker, Kubernetes (EKS), and GitHub Actions, supporting rapid scaling and reliability across cloud and hybrid environments.
- Integrated AWS services including API Gateway, S3, RDS, DynamoDB, and Lambda for secure, high-performance cloud applications.
- Led configuration, SCM, and build/release management, focusing on optimizing deployment and monitoring of diagnostic tooling.
- Conducted user research and usability testing, implementing advanced troubleshooting and customer satisfaction improvements.
- Drove code quality and maintainability through code reviews, developer mentoring, and consistent adherence to SDLC best practices.

Conquer Technologies – India

Software Developer

May 2019 – Feb 2021

- Architected and developed microservices focused on diagnostics and business logic layers, utilizing Spring Boot, Spring Cloud, Docker, Kubernetes, and Hibernate ORM frameworks.
- Wrote and maintained complex database access layers, optimizing query performance for large-scale data products.
- Implemented BDD scripts and automated acceptance testing, enabling behavior-driven diagnostics for quality-controlled process environments.
- Built dashboards and web interfaces in ReactJS and Bootstrap, improving end-user experience and system performance.
- Provided technical leadership for dependency injection, distributed tracing, and integration patterns using Kafka and RabbitMQ.

Projects:

FurniReality – Augmented Reality for E-Commerce Furniture Shopping

- FurniReality is a web-based application that enables users to visualize and select furniture by placing 3D models in their homes using AR technology via web cameras or mobile devices.
- The project bridges digital catalogs with physical environments to enhance e-commerce experience and reduce product returns.
- Developed diagnostic service APIs for real-time data validation, error logging, and device compatibility, enabling robust system monitoring and analytics dashboards for user interactions.
- Led the deployment of scalable backend infrastructure with Docker, Kubernetes, and AWS, and ensured secure transactions through OAuth2 and JWT authentication.

Technologies Used: Angular, React, Java, Spring Boot, WebAR, Three.js, Python, REST APIs, AWS S3, Docker, Kubernetes, MongoDB, JWT, OAuth2, Jenkins, Postman.

New York Housing Price Prediction Pipeline

- Developed a predictive analytics pipeline to estimate residential housing prices in New York City using raw real estate data. The solution incorporated data ingestion, cleansing, exploratory analysis, advanced feature engineering, and regression model training for accurate price forecasting.
- Diagnostic logging and error-handling routines were built to track data quality and model transparency. Results were served through REST APIs and visualized on interactive dashboards.
- Contributed to generative AI model integration for improved accuracy and observability, collaborating with data scientists on continuous product enhancement.

Technologies Used: Python, Pandas, NumPy, Scikit-learn, Java, Spring Boot, Angular, REST APIs, AWS RDS, Docker, Kubernetes, Jenkins, Swagger, Postman, ELK Stack.
